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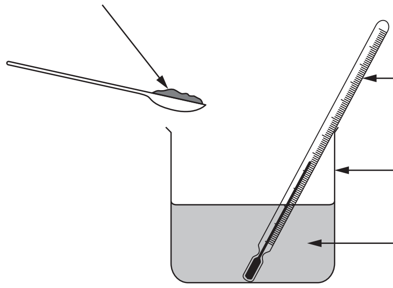
3. Magnesium, zinc and iron powders were each added separately to 100 cm^3 of copper(II) sulfate solution, to see which gave the greatest temperature change.



The temperature was recorded before and after each reaction. The results are shown in the table.

Metal	Temperature before the reaction (°C)	Temperature after the reaction (°C)	Temperature increase (°C)
zinc	20	....	14
magnesium	19	39	20
iron	19	24	5

(a) Calculate the temperature after the reaction with zinc. [1]

Temperature = ... °C

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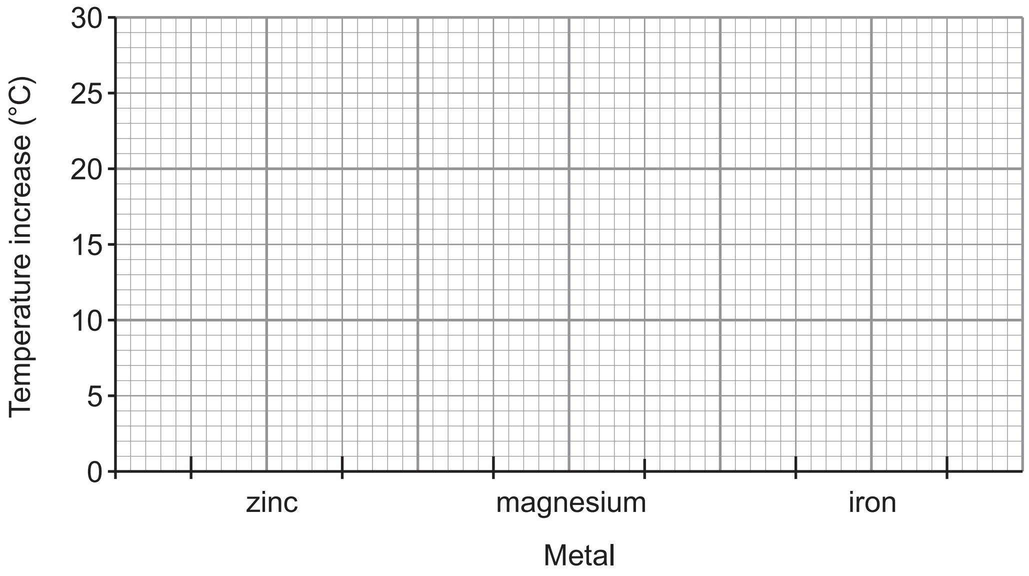
(b) Plot a **bar chart** to show the temperature increase for each metal. [2]



(c) (i) Calculate the energy released in the reaction with magnesium. [2]
Use the following equation.

energy released (J) = volume of solution (cm^3) × 4.2 × temperature increase (°C)

Energy released = ... J

(ii) In this reaction magnesium sulfate, $MgSO_4$, is formed. What is the relative formula mass (M_r) of magnesium sulfate? [2]

A_r(Mg) = 24 A_r(S) = 32 A_r(O) = 16

M_r = ...

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7